

Software in This Month's DVD

This month's EFY Plus DVD brings you a useful FPGA CAD tool, tools for embedded software development and VHDL simulation, FSM designer and a timeline diagram editor for your utilities collection

PANKAJ V.

Weka

Methods of machine learning help in analysing large data and decide about the most relevant information. Weka provides you with a collection of such machine learning algorithms for solving real-world data mining problems.

This tool is written in Java and can be run on any platform. You can directly apply these algorithms to a dataset or use your own Java codes to call them. Weka is well suited for developing new machine learning schemes and contains tools for data pre-processing, classification, regression, clustering, association rules and visualisation. **Supported OS:** Windows, Linux, MacOSX **Link:** <http://sourceforge.net/projects/weka/files/>

BERTOS

BERTOS is a real-time open source operating system that comes with drivers and libraries designed for the rapid development of embedded software. BERTOS is a free tool with no licence costs or royalty, allowing you to cut the economic investment for your products. This multi-platform tool can be useful for building commercial applications with its modular design, which allows running it on different architectures, ranging from tiny 8-bit microcontrollers such as the Atmel AVR to the 32-bit ARM architecture.

Its pre-emptive multitasking kernel implements many IPC primitives. And there is a hardware abstraction layer which includes a large number of peripheral drivers (timer, serial, ADC, motors, LCD display, NTC sensors, keyboard, buzzer, memories), algorithms (hash table, CRC, MD2, entropy pool, RLE), communication protocols

and a graphic windowing sub-system for small displays. **Supported OS:** Windows, Linux, MacOSX **Link:** <http://download.bertos.org/release/>

GHDL

GHDL is an open source VHDL simulator. It is not a synthesis program, that is, one cannot create a netlist using this tool but it can execute almost any VHDL program. It uses the GCC at the back end and directly translates a VHDL file to machine code without using an intermediary language such as C or C++. **Supported OS:** Windows, Linux **Link:** ghdl.free.fr/

KJWaves

KJWaves is a cross platform SPICE tool written in Java. You can have easy modification and viewing of SPICE circuit files with this program. This program can read SPICE3 RAW format (as well as GnuCap) and create graphs of results through an interactive GUI with features such as support for waveform math, copying to clipboard and saving. **Supported OS:** Windows, Linux **Link:** <http://sourceforge.net/projects/kjwaves/>

QFSM

Finite state machines (FSMs) are a model to describe complex objects or systems and can be used to design integrated circuits or to create regular expressions, scanners or other program code. QFSM is a graphical editor for FSMs which is written in C++. It can be used for easy drawing, editing and printing of diagrams. You can create your own VHDL test codes and have interactive simulations using QFSM. It supports HDL exports in different file formats, such as, AHDL, VHDL, Verilog HDL, KISS

and diagram exports in EPS, SVG and PNG formats. **Supported OS:** Windows, Linux **Link:** <http://sourceforge.net/projects/qfsm/>

RapidSmith

RapidSmith is an open source EDA tool. This FPGA CAD tool framework is written in Java for modern Xilinx FPGAs. It is based on XDL (Xilinx design language) and provides interactive graphical tools for Xilinx XDL/NCD designs. It can be used as a rapid prototyping platform for research ideas and algorithms relating to low-level FPGA CAD tools. **Supported OS:** Windows, Linux, MacOSX **Link:** <http://sourceforge.net/projects/rapidsmith/>

Timing Editor

Timing editor is a simple tool for drawing and editing timeline diagrams. It allows you to draw signals, clocks and busses, set length of a tick and number of displayed ticks, set transition in percentage of a tick and also set time compressions. This windows-based easy-to-use tool also allows you to undo or redo your steps, save and open your diagrams and export them to PNG, SVA or PS formats. **Supported OS:** Windows **Link:** <http://sourceforge.net/projects/timingeditor>

PIC Simulator IDE 6.96

PIC Simulator IDE is a powerful application that supplies PIC developers with user-friendly graphical development environment for Windows with integrated simulator (emulator), basic compiler, assembler, dis-assembler and debugger. **Supported OS:** Windows **Link:** <http://www.oshonsoft.com/downloads/page.php> ●

The author is a technical journalist at EFY, Gurgaon